Aaron Espinoza

Austin, TX | 815-503-1233 | me@acespinoza.net
acespinoza.net | github.com/Acesp25 | linkedin.com/in/aceespinoza

SYSTEMS EXPERIENCE

Open Source Contributor **Sept. 2025 – Present**
The FreeBSD Project *Remote*

- Fixed a kernel panic in `if_bridge(4)`: a NULL `softc` dereference on the mbuf pullup failure path in `bridge_input()`. Found while tracing the driver during early `veb(4)` work; reviewed and landed through the project's code review process. Commit history
- Developing `veb(4)`, a virtual Ethernet bridge pseudo-device built on `if_clone(9)` and the `ifnet` KPI: stripped down relative to `if_bridge(4)` and hidden from the host network stack, designed to cut per-packet overhead for latency-sensitive jail and bhyve networking.

Google Summer of Code Contributor **May 2025 – Sept. 2025**
The FreeBSD Project *Remote*

- Delivered a modular Rust framework for FreeBSD kernel device drivers with safe wrappers over C kernel objects (devfs nodes, character devices, memory allocation), removing the need for contributors to hand-roll FFI per driver.
- Wrote 11 automated ATF/Kyua tests covering stress, concurrency, error paths, and memory leaks; integrated into CI so regressions surface on every change.
- Fixed devfs registration, quiesce/unload handling, and LLVM 19 compatibility in existing Rust drivers, restoring a stale build against `CURRENT`; shipped reference drivers and documentation now used by new contributors.
- Project writeup | Official FreeBSD project video

Maintenance Technician **Aug. 2022 – May 2024**
tekRESCUE *San Marcos, TX*

- Ran weekly maintenance across client Windows and Linux fleets; specified and built custom servers and handled on-site hardware and network troubleshooting.

SYSTEMS PROJECTS

veb(4) — Virtual Ethernet Bridge for FreeBSD In progress
Inspired by OpenBSD's Layer 2 bridge pseudo-device, but for the FreeBSD network stack. It is an optimized path for L2 bridging built with ifnet(9) and netlink(4). *C, FreeBSD, ifnet, netlink, ATF/Kyua*

RustKLD — Rust Kernel Driver Framework for FreeBSD github.com/Acesp25/rustkld
Modular framework and test harness for FreeBSD kernel loadable drivers written in Rust: safe abstractions over C kernel interfaces, CI integration, and contributor documentation. *Rust, C, FreeBSD, ATF/Kyua.*

Homelab Ongoing
Built and operate three servers: a FreeBSD pf router with filtered DNS; a FreeBSD host running bhyve VMs as kernel build/test targets and jails for services, with NFS over WireGuard and daily ZFS snapshots; and an Ubuntu RTX 3060 node for training GAN models for research and running local LLMs. *FreeBSD, Ubuntu, pf, WireGuard, bhyve, jails, ZFS, CUDA, PyTorch*

RESEARCH & PUBLICATION

Undergraduate Deep Learning Researcher, Texas State University **Aug. 2024 – Present**

- GAN-based adversarial perturbation research against facial recognition; 98.7% average impersonation success on LFW and CASIA-WebFace.
- Placed 1st globally in the MICCAI 2024 WHS++ segmentation challenge as an AI research intern at Guangdong Provincial People's Hospital (May–Aug. 2024).
- *Enhance Multi-modal and Multi-center Whole Heart Segmentation Using Data Augmentation and Model Calibration.* CARE Workshop, MICCAI 2024. doi.org/10.1007/978-3-031-87009-5_13

EDUCATION

Texas State University **Expected Dec. 2026**
B.S. Honors Computer Science; Minors in Applied Mathematics and Chinese *San Marcos, TX*

- GPA 3.98/4.00 overall, 4.00/4.00 College of Science & Engineering GPA. Dean's List/President's List, all semesters.
- Relevant coursework: Computer Architecture, Parallel Programming, Operating Systems, Computer System Fundamentals.
- Extracurriculars: Vice President, TXST Fencing Club (40+ members), TXST Esports Club CS2 Captain.

SKILLS

Systems: FreeBSD kernel and device drivers, KLDs, devfs, VirtIO, network pseudo-devices, POSIX (pthreads, sockets, IPC)
Languages: C, Rust, Python, Bash, Assembly (MIPS, x86-64), Java, C++
Networking: Ethernet L2 Bridging, TCP/IP, `pf`, WireGuard, NFS, unbound
Virtualization & Tooling: bhyve, QEMU, jails, ATF/Kyua, DTrace, GDB, Wireshark, CI pipelines, Git, Vim, VS Code
ML: PyTorch, CUDA, TensorFlow, Model training